

1. **Scenario:** A system checks if a user is eligible to vote based on their age.

Answer:

1. Get input(age) from the user.
2. Check if age is greater than or equal to 18.
3. If yes then print Eligible to vote.
4. If not then print not eligible.

2. **Scenario:** A program processes a list of numbers and needs to find the largest value.

Answer:

1. Analyse a list of numbers.
2. Define a function to call 1st number in the list.
3. Compare the other numbers in the list with current largest number.
4. If it is greater than the other number.
5. Then return largest number.

3. **Scenario:** A company provides employees with a 10% bonus if their salary exceeds \$50,000.

Answer:

1. Get the salary details.
2. If salary is greater than 50000 then calculate 10% bonus.
3. Print the calculated bonus amount.
4. Otherwise print no bonus.

4. **Scenario:** A program evaluates a number to determine if it is even or odd.

Answer:

1. Enter the number to check.
2. If the number is divisible by 2.
3. Then print even.
4. Else print odd.

5. **Scenario:** A text-processing tool reverses a given word or sentence for formatting purposes.

Answer:

1. Get the input from the user.
2. Define a function to reverse the word.
3. Use the reverse python slicing to reverse the word.
4. Return the reversed word.

6. **Scenario:** A grading system determines whether a student has passed or failed based on their score.

Answer:

1. Get the score from the user.
2. Check the score is greater than or equal to 40.

3. If yes then print Pass.
4. Otherwise, print Fail.
7. **Scenario:** A retail store offers a 20% discount if a customer's total order exceeds \$100.
Answer:
 1. Get the total amount.
 2. Check whether the total amount is greater than 100.
 3. If yes, then calculate 20% discount and subtract from the total amount.
 4. Return the final amount.
8. **Scenario:** A banking system processes withdrawal requests and ensures the user has enough balance.
Answer:
 1. Get the balance amount.
 2. Get the amount to withdraw.
 3. If the amount is less than or equal to the balance amount.
 4. Subtract the withdraw amount from the balance amount.
 5. Then print Withdrawal successful and remaining balance
 6. Otherwise print insufficient balance.
9. **Scenario:** A calendar system verifies whether a given year is a leap year based on standard leap year rules.
Answer:
 1. Enter the year to check.
 2. Check it is divisible by 400.
 3. Or check it is divisible by 4 and not divisible by 100.
 4. If it is yes, then it is a leap year.
 5. If no, then it is not a leap year.
10. **Scenario:** A program filters out only even numbers from a given list.
Answer:
 1. Read the list of numbers.
 2. Create an empty to list.
 3. Check the numbers in the list is divisible by 2.
 4. If true then add the even numbers to empty list.
 5. Return the list containing the even numbers.